Frontier



Arca

Carbon dioxide removal prepurchase application
2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

Arca Climate Technologies Inc.

Organization location (we welcome applicants from anywhere in the world)

Vancouver, British Columbia, Canada

Name(s) of primary point(s) of contact for this application

Artem Bocharov
Paul Needham

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

For-profit Canadian corporation

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

Project AMI (Activation Minérale Industrielle)

b. **Description of the CDR approach:** Describe how the proposed technology removes CO_2 from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us how your project addresses any of the priority innovation areas identified in the RFP. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

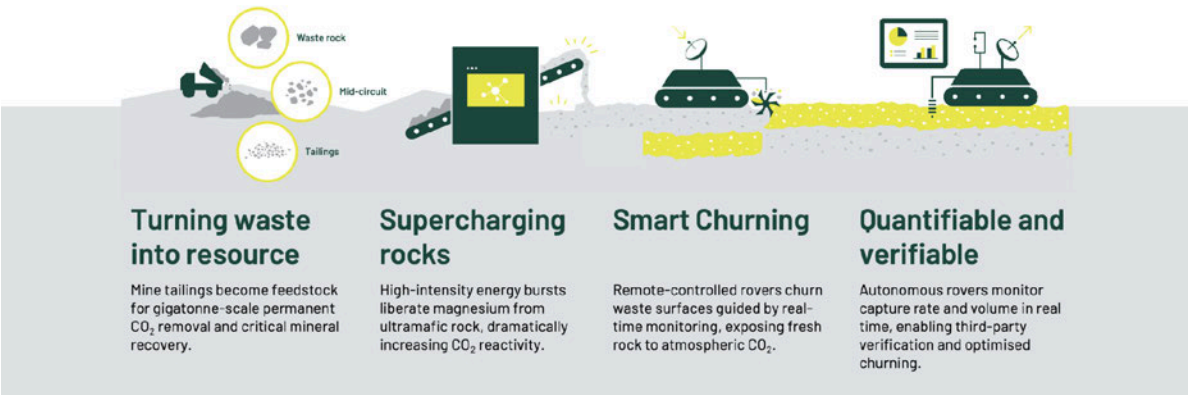
Earth has been pulling CO_2 out of the atmosphere and turning it into rock for billions of years. The conversion of atmospheric CO_2 into solid carbonate minerals is a fundamental Earth process that stores carbon in stable rock over geological timescales. When magnesium-rich rocks are exposed to air and moisture, they react with atmospheric CO_2 to form magnesium carbonate minerals, new rocks that permanently store CO_2. This natural geochemical process is called carbon mineralization. Under ambient conditions, these reactions proceed very slowly, often over millennia, limiting their relevance for near-term climate mitigation.

Arca has developed technologies and processes to accurately measure – and accelerate – carbon mineralization at scale. Our feedstock is magnesium-rich mine tailings: finely ground ultramafic rock that is the byproduct and waste stream from the mining of critical metals such as nickel. There are already billions of tonnes of this material on the surface of the earth from historical mining, representing a large potential resource for industrial-scale carbon removal. Arca has developed an innovative process using microwave energy to supercharge the ability of this material to capture and permanently store carbon dioxide.

In short, Arca repurposes historical mine waste for durable, measurable, industrial-scale carbon dioxide removal. We call it industrial mineralization, and it has the potential to permanently remove gigatonnes of CO_2 from the atmosphere.

Figure 1: Integrated Field Deployment Model

[1] We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your organization.



How the Carbon is Removed and Stored

Arca's technologies focus on processing serpentinite-rich mine tailings, which contain abundant magnesium silicate minerals, but in their natural state these react with CO_2 far too slowly to be relevant to the current climate challenge. Our Mineral Activation system uses a continuous-flow microwave process that selectively targets serpentine, disrupting its crystal structure through dehydroxylation, a process that removes water molecules and reconfigures the mineral lattice into a disordered, highly reactive form. The result is material capable of achieving greater than 20 weight % CO_2 uptake under ambient conditions over just several weeks, compared to near-zero uptake from unactivated feedstock over the same period. In other words, Mineral Activation enables Arca to sequester over 1 tonne of CO_2 in every 5 tonnes of mine waste.

Once activated, the mine tailings are spread across the storage facility, where Smart Churning takes over. By agitating the tailings, we increase the surface area, and therefore the amount of magnesium exposed to CO_2 in air that it can react with. This process also optimizes moisture content in the tailings, the second crucial factor in the speed of the mineralization reaction. Rather than simply plowing the area on a fixed schedule, Smart Churning uses continuous real-time measurement of CO_2 flux (the rate at which CO_2 is moving from the atmosphere into the tailings) to determine precisely where and when to churn the material to achieve the fastest reaction, based on weather, available moisture, surface passivation, and other factors. The system churns only when and where the data says it will drive additional reaction, which makes it inherently efficient. Arca's pilot deployments at a large nickel mine in Western Australia in 2024–25 demonstrated significant increases in reaction rates relative to baseline passive mineralization.

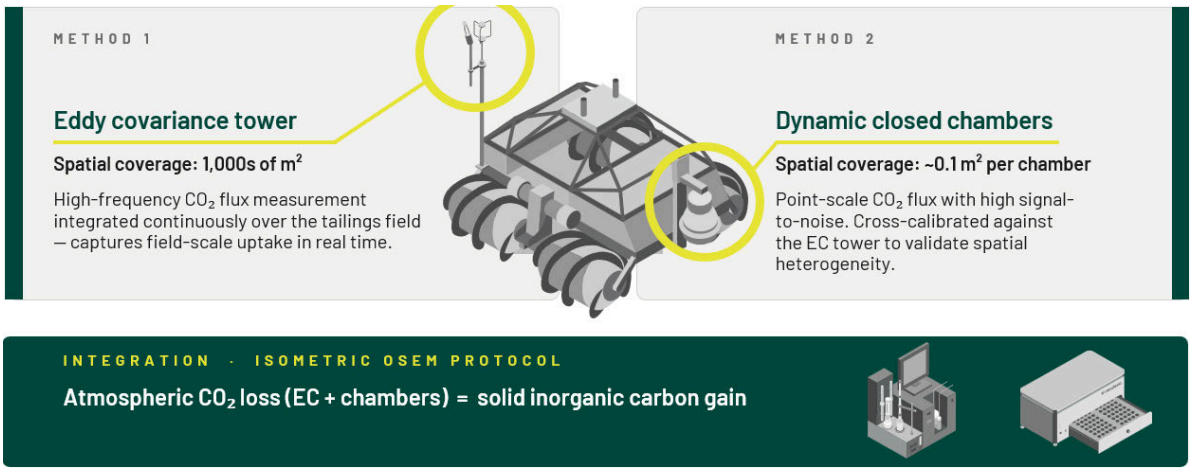
As CO_2 is absorbed, it combines with magnesium released from the activated mineral to precipitate hydrated magnesium carbonates, such as hydromagnesite, which form within the pore spaces of the tailings. CO_2 is chemically bound within a carbonate crystal lattice and remains stable on geological timescales of 10,000+ years. The risk of reversal is negligible: releasing the CO_2 would require extremely high temperatures or exposure to strong acids - conditions that do not naturally exist at these mine tailings facilities. The storage is permanent and requires no ongoing management.

Measurement, Reporting, and Verification

Arca's MRV system directly measures the movement of CO_2 from the atmosphere into mine tailings using two independent, co-deployed methods: eddy covariance towers, which track CO_2 movement between the air and the tailings over large areas, and dynamic closed chambers, which provide high-resolution localized measurements of CO_2 uptake for calibration and spatial validation. Both

systems use commercially available equipment, calibrated for Arca's purposes, and apply well-understood scientific techniques to generate accurate, verifiable MRV data.

Figure 2: Direct, Continuous, Redundant CO2 Flux Measurement and Verification



Unlike many other CDR pathways, Arca's MRV approach does not rely on models, proxies, or stoichiometric estimates. It directly tracks CO_2 flux across the actual deployment area. Mine tailings are a relatively simple measurement environment: unlike marine bodies and agricultural soils, they are homogenous and contain negligible organic carbon and biological activity, eliminating the largest sources of measurement ambiguity in those systems. This makes MRV easier, more transparent, and more precise.

Differentiation and Competitive Position

Arca's approach is significantly differentiated from other CDR pathways:

1. **Arca creates reactive feedstock at scale.** Serpentinite mine tailings are among the most abundant magnesium-rich materials on Earth — tens of billions of tonnes of waste material, already ground, already on the surface, already at scale. However un-activated and unprocessed tailings capture very little CO_2 because most of the material is buried and the minerals are stable and unreactive. Arca's Mineral Activation process changes that. By passing the material through a continuous-flow microwave system, Arca disrupts the crystal structure of serpentine, transforming an otherwise inert waste into a material capable of capturing more than 20 wt% CO_2 under ambient conditions — a ratio that enables one tonne of CO_2 sequestered per five tonnes of mine waste, with no chemical reagents and no pressure vessels.

2. **MRV that measures rather than models.** Most open-system CDR pathways rely on stoichiometric estimates, geochemical proxies, or modelled flux partitioning. Arca directly measures CO_2 movement from the atmosphere into the tailings using co-deployed eddy covariance towers and dynamic flux chambers — the same instrumentation stack used in atmospheric science, applied to the simpler measurement environment of mine tailings (homogeneous, low organic carbon, negligible biological activity). Flux measurements are further validated with periodic measurement of mineralized CO_2 using XRD, Raman Spectroscopy and/or Coulometric analysis. This approach has been demonstrated and independently validated: TRL 8.

3. **CO_2 stored as rock, permanently.** Carbon mineralization stores CO_2 as solid magnesium carbonate minerals — hydromagnesite and nesquehonite — chemically bound within a crystal lattice. Releasing that CO_2 would require temperatures above 350°C or exposure to strong acids. Neither condition naturally exists at a tailings storage facility. The storage requires no costly ongoing management, no costly monitoring for leakage, and no additional containment infrastructure. Field evidence from natural hydromagnesite deposits — including playas in British Columbia that have been accumulating carbonate minerals since the last glacial retreat, over 10,000 years ago — confirms that these minerals are stable under surface conditions on geological timescales.

4. **No new land.** Arca's industrial mineralization pathway works within the existing industrial footprints of the mines. By avoiding incremental habitat disturbance and competing land uses, this approach reduces permitting friction, accelerates deployment timelines, and structurally lowers total program cost and community impact risk relative to CDR projects that depend on dedicated land banks or new infrastructure corridors

5. **Remediation is built in.** At legacy asbestos mine tailings sites, Mineral Activation does something passive mineralization cannot: it destroys chrysotile fibres. Third-party testing confirms fibre destruction to below detection limits following activation. Processing tailings for CDR and simultaneously destroying fibres that represent a long-term health hazard at the site is an important co-benefit for host communities.

6. **Critical metal recovery, at some sites.** Mineral Activation also disrupts the silicate matrix that locks residual metals like nickel, cobalt, and others within the tailings. At sites where these metals are present in recoverable concentrations, this creates a pathway to separate and sell critical metals as a co-product of CDR production. Minimal additional process steps are required beyond what the activation system already does. The economics vary by site and metal concentration, but at nickel-bearing tailings the potential offset is material. Arca has demonstrated metal liberation at the bench scale and is developing the downstream recovery process. Where it applies, it changes the cost structure of CDR in a way no other mineralization approach can replicate.

Alignment with Frontier's 2026 Innovation Areas

This project directly addresses three of Frontier's actively solicited innovation areas.

Enhancing surficial mineralization. Arca enhances surficial mineralization by combining feedstock activation with field-scale process control to increase reaction rates. Mineral Activation is a feedstock pretreatment technique that enhances reactivity without requiring high-pressure reactors or continuous chemical inputs. Smart Churning offers a simple, efficient mineralization process that optimizes the transformation of a greenhouse gas into durable solid form in real-world tailings environments. Deployed together and in sequence, these technologies address the core tradeoff Frontier identifies: balancing CapEx and OpEx to maximize feedstock reactivity.

Improving the MRV toolkit for open-system pathways. Arca's dual eddy covariance and dynamic chamber platform is designed specifically for the demands of open-system CDR at field scale. It provides continuous, high-fidelity atmospheric flux measurement and is at TRL 8 following a successful pilot in 2024–25. By combining scalable instrumentation with direct CO2 flux measurement, the system reduces reliance on labor-intensive sampling and model-based inference

while maintaining quantification accuracy.

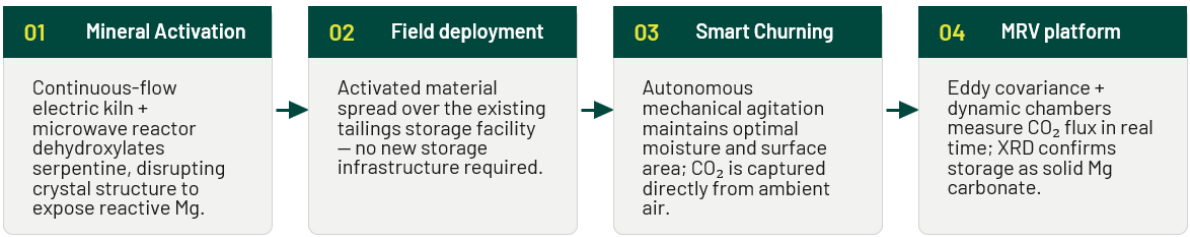
Novel approaches with breakthrough potential. Arca's system introduces a step-change improvement in mineralization performance by fundamentally increasing feedstock reactivity and enabling real-time process optimization. In addition, Mineral Activation liberates residual nickel from tailings, creating a pathway for value recovery that can offset costs. Arca is developing a process to separate and concentrate this liberated nickel for downstream sale, with revenues supporting the cost reduction journey toward <$100/t$CO_2$. Combined with a feedstock base of tens of gigatonnes of existing global serpentinite tailings, plus billions of additional tonnes generated annually from active mining, this positions Arca's platform for gigatonne-scale CDR at costs well below Frontier's long-term targets.

c. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale criteria?[2] What is your approach to quantifying the carbon removed? Please include figures and system schematics, and be specific but concise. 1000-1500 words

Under the Business as Usual (BAU) scenario, the legacy mine waste material is stockpiled, mostly buried, not exposed to air and effectively inert. Our objective is to demonstrate that Arca's end-to-end process, including material handling, Mineral Activation plus Smart Churning increases the rate and total capture by several orders of magnitude over BAU.

The primary objective of Project AMI is to design, build, and operate an integrated mineral activation and carbon mineralization system that converts atmospheric CO_2 into stable magnesium carbonate minerals within an existing mine footprint. The project will demonstrate continuous mineral activation, field deployment of activated material, and real-time quantification of CO_2 removal using Arca's MRV system. Together, these components form an end-to-end system linking mineral activation, CO_2 capture and mineralization, and direct quantification under real-world operating conditions.

Figure 3: Arca's Industrial Mineralization Process.



Location: Project AMI will run at a legacy mine in the town of Thetford Mines, QC, Canada. The site

[2] We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at $100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.

holds approximately 66 Mt of serpentinite tailings that were previously crushed and ground during asbestos extraction, providing access to large volumes of suitable feedstock within an existing industrial footprint and without the need for new storage infrastructure. The specific mine site is one of several in the area, where there are more than 450 million tonnes of waste from historical mining.

This project will use a Mineral Activation unit, a Smart Churning system, and a multi-scale MRV platform to generate CDR.

Scale: This project has a gross nameplate capacity of 3,035 tCO_2/yr. At full capacity, Arca will process approximately 11,800 tonnes of rock per year, with mineralization occurring over tens of hectares of the existing storage facility, delivering

Duration: This project runs for 3 years from 2028 to 2030

Expansion opportunities:

The first three years of operation of this project until end of 2030 will validate the Industrial Mineralization pathway, and the combined Mineral Activation, Smart Churning, and MRV systems that Arca will deploy. From 2030 Arca will initiate an expansion of operations, scaling to processing approximately 1 million tonnes of rock per year – comparable in size to a small mining operation. CapEx investment for this project will transition and continue to be utilized by the larger FOAK project, which will deliver ~ 250,000t of net CO_2 removed per year. The 66 Mt tailings resource at this one project site is sufficient to sustain commercial-scale CDR production for decades. Nearby sites contain about 450 million tonnes of suitable feedstock, providing opportunities for significant scaling up.

Costs: During this project (2028-2030), the cost of CO_2 removal is driven primarily by labor (“Fixed OpEx” in the TEA). Each piece of equipment requires a minimum operating crew regardless of throughput, making labor costs disproportionately high relative to the volume of CO_2 removed at this demonstration scale. The remaining cost breakdown is split approximately equally between capex recovery, equipment maintenance, and electricity. The vast majority of MRV cost is the one-time upfront cost of equipment and is therefore included in capex costs.

Labor intensity at this scale is a well-understood characteristic of pilot-scale industrial operations, not a technology-specific risk. As the system scales to commercial throughput (1 Mt rock/yr), labor cost per tonne of CO_2 removed falls sharply through automation of material handling, remote monitoring, and process control — the same optimization pathway followed by every continuous industrial process moving from pilot to full scale.

Reaching Frontier's cost and scale targets requires two parallel improvements: economies of scale from increased throughput, and increased reactivity of the feedstock. Scaling throughput reduces labor, capex recovery, and equipment maintenance costs on a per-tonne-of-CO_2 basis. Increasing reactivity — lowering the mass of rock that must be processed per tonne of CO_2 removed — reduces total installed power requirements and therefore both opex and capex. Electricity costs will decrease sharply as heat recovery is implemented at progressively higher TRLs. Together, these drivers provide a clear and well-precedented pathway to low-cost, high-throughput carbon removal at the scale Frontier is targeting.

MRV: Carbon removal will be quantified using processes already demonstrated at a previous project and third-party verified (TRL 8). Arca uses a direct, measurement-based approach that captures CO_2 flux from the atmosphere into the tailings. Eddy covariance towers provide continuous, area-integrated measurements over hectare-scale footprints, while dynamic closed chambers provide

localized, high-resolution flux measurements for calibration and validation. By combining independent measurement approaches across spatial scales, the system enables high-confidence quantification of CO_2 removal without reliance on modelled or proxy-based estimates. This approach directly links atmospheric CO_2 uptake to mineral storage and provides a robust basis for verification at scale.

The secondary objective of this project is to validate, at scale, the remediation benefits of Mineral Activation. Third-party analysis of our activated material indicates that our Mineral Activation process destroys asbestos fibres – often to levels below detection limits. If validated at larger scale, this would be an essential community benefit of Arca's Mineral Activation CDR projects using asbestos tailings.

In the Thetford region, the harm from legacy asbestos tailings is a long-running, chronic exposure issue. Quebec public-health analyses found asbestos fibres in both indoor and outdoor air in Thetford Mines and concluded that those levels imply a measurable long-term health risk for residents. Research in the Bécancour watershed likewise shows that erosion from historic mine wastes in the Thetford area still releases asbestos fibres, with contamination persisting in downstream sediments and waterways. Deploying Mineral Activation system would not erase the region's asbestos legacy overnight, but destroying fibres in material that is already being processed for CDR production has the potential to alleviate windblown dust and fibre mobilization and supporting eventual remediation of the site.

d. **Risks:** What are the biggest risks and how will you mitigate those? Include technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. 500-1000 words

Technical: The primary technical risk is reactor scale-up: moving the Mineral Activation system from TRL 5 to TRL 7/8 requires demonstrating continuous-flow operation at pilot scale with uniform heat distribution, controlled waveguide design, and hot-spot management across a larger processing volume. A secondary risk is feedstock mineralogy variability: chrysotile, lizardite, and antigorite respond differently to microwave activation, and while the waste is predominantly chrysotile, spatial variation in polymorph ratios could affect consistency of reactivity at scale. While activation parameters (e.g., time-temperature profile) must be adjusted across these phases and with varying Fe content, we have not observed significant differences in overall energy requirements. Given these effects are well-characterized and do not materially impact energy requirements, the remaining risk is further mitigated through a staged scale-up pathway, a proprietary activation protocol library covering more than ten ultramafic compositions, site-specific testing currently underway, and a CDR delivery start date of 2028 that provides a meaningful development buffer.

Arca has been operating a field-deployable flow-through Mineral Activation reactor at its R&D facility, processing material supplied by multiple partners. Arca will next conduct a field trial of Mineral Activation at a site local to the company’s headquarters. The site contains asbestos-bearing material similar to that of target project site in the Thetford region. This pilot will run for 9+ months to inform the final design, commissioning, and operation of the system deployed at Project AMI.

Project Execution: The primary execution risk is site access, as a formal agreement with the site owner is not yet executed. Heads of terms with the site owners are currently verbal; a formal legal agreement is targeted for H2 2026. With respect to permitting, the target project site has existing mining permits held by the owner. Based on preliminary discussions, project activities conducted

within the existing mine footprint may not require new environmental authorizations. However, it is possible that certain project activities, such as scale increases, could require authorizations under Quebec's Loi sur la qualité de l'environnement or coordination with Ministère de la Santé et des Services sociaux (MSSS) on asbestos specific operational requirements, introducing some schedule uncertainty. Deployment also carries procurement and mobilization risk that could push first CDR delivery into the latter part of 2028. These risks are mitigated by strong community and municipal support in Thetford Mines, including the Du Gris au Vert initiative and the municipal economic development office, which creates favorable conditions for regulatory engagement. Additional nearby sites in Quebec provide optionality if the first location proves unexpectedly challenging to deploy at. Arca has engaged an engineering firm to lead design development and regulatory pathway definition.

MRV: Pre-existing carbonate content in the tailings is spatially heterogeneous, introducing baseline uncertainty that requires spatial sampling before carbonation begins. Isometric's OSEM protocol is the leading MRV standard for enhanced weathering and mineral carbonation, but its application at FOAK field deployment scale is relatively new, and audit findings may require adaptive measurement adjustments during operation. Mitigations include a complementary measurement stack —24/7 EC tower and dynamic closed flux chambers, complimented by less frequent solid-phase analysis.

Ecosystem: Handling and thermally activating chrysotile-containing tailings generates potential for fibre aerosol. Although Mineral Activation progressively destroys chrysotile fibre structure — and fibre destruction is a stated co-benefit of the process — dust suppression during material handling and Smart Churning requires active management. High-temperature processing also requires exhaust management and local air quality monitoring. Mitigations include an enclosed Mineral Activation processing unit, real-time air quality monitoring at the site boundary, Smart Churning system design that integrates water management, monitoring of fibre destruction progression at defined intervals, and proactive regulatory engagement with MSSS on asbestos-specific operational requirements.

Financial: The primary financial risk is securing sufficient capital to execute the project through deployment and early operations. While near-term funding is supported by existing balance sheet resources and committed grant funding, additional equity and/or grant financing will be required to fully deliver the project. There is also longer-term cost risk if expected reductions from scale-up, throughput increases, and Smart Churning automation are not realized as planned. These risks are mitigated through multiple channels. The Frontier pre-purchase provides non-dilutive revenue that supports the near-term funding gap. In parallel, a signed multi-year corporate offtake agreement provides forward revenue visibility beyond this contract, reducing dependence on future market conditions. Capital deployment is staged and aligned with technical milestones, limiting exposure ahead of key validation points and allowing for disciplined scaling as performance is demonstrated.

Indigenous engagement: Indigenous engagement and regulatory clarity represent additional risk considerations. We are engaging Mokwateh – Canada's leading Indigenous Relations consulting firm to support our work. Mokwateh mapping of Indigenous territorial claims in the Thetford Mines region has been commissioned but is not yet complete. If claims are identified, free, prior, and informed consent processes will be required before project commencement; this is a schedule dependency that Arca is addressing proactively rather than an anticipated obstacle.

Reputational / stakeholder perception: Reputational sensitivity around the "asbestos" label is real in public communications and buyer due diligence contexts; Arca manages this by centering the narrative on fibre destruction — Mineral Activation converts a legacy liability into permanent CO_2

storage.

Regulatory pathway: Regulatory frameworks for CDR involving asbestos-bearing materials are still evolving in Quebec, creating some uncertainty in permitting requirements and timelines. Arca is engaging proactively with MELCCFP and MSSS to ensure the regulatory pathway is clearly defined well in advance of deployment.

e. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) *(Should be net volume after taking into account the uncertainty discount proposed in 5c)*	1,500 tonnes
Delivery window *(At what point should Frontier consider your contract complete? Should match 2f)*	December 2030
Levelized cost ($/ton CO_2) *(This is the cost per ton for the project tonnage described above, and should match 6d)*	$833
Levelized price ($/ton CO_2)[3] *(This is the price per ton of your offer to us for the tonnage described above)*	$1000

[3] This does not need to exactly match the cost calculated for "This Project" in the TEA spreadsheet (e.g., it's expected to include a margin and reflect reductions from co-product revenue if applicable).